

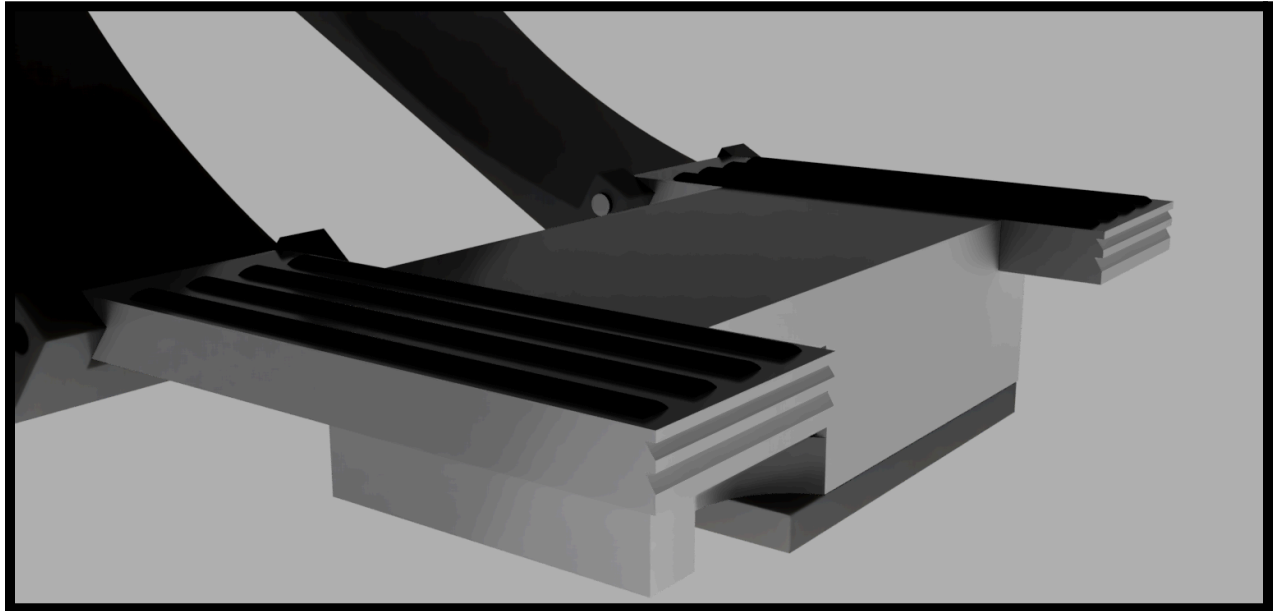
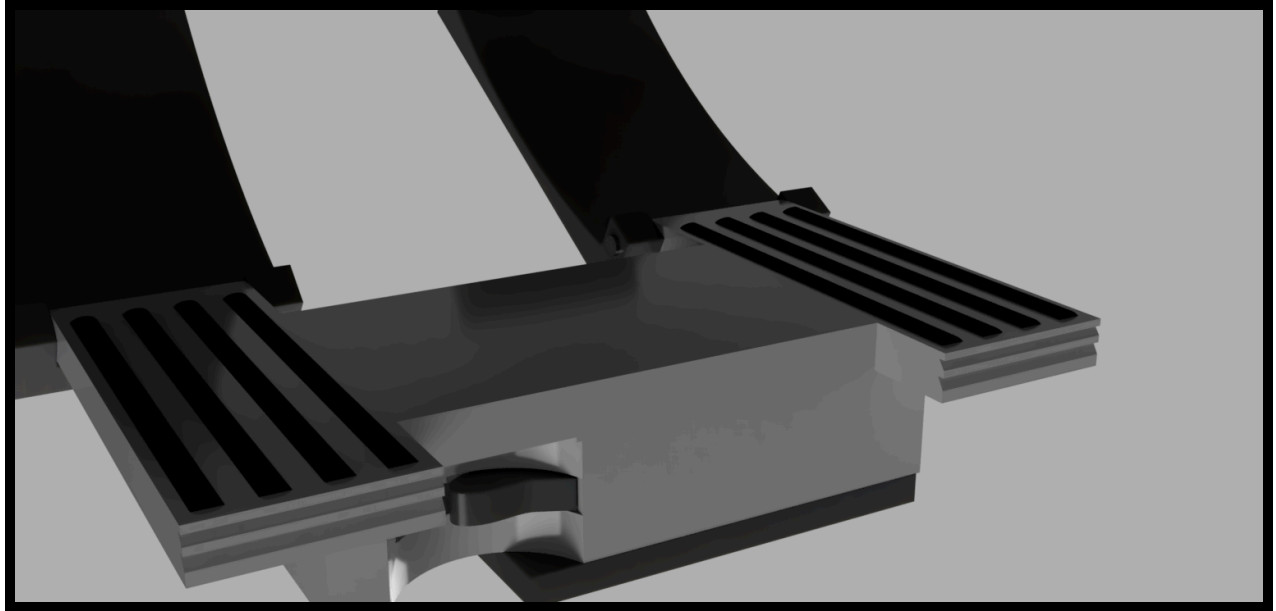
HydroClip Final Report

By: Aahil Ali

Designing for manufacturing

The Hydroclip has undergone several variations, and I have opted for a few design changes as well as choosing different manufacturing techniques than the original prototype. The design for HydroClip consists of the following custom items: the water bottle holder, the buckle insert, the buckle receiver, the mounting plate, and the mounting clamps. Because all of these pieces have unique three-dimensional geometries, I have opted to use injection molded PolyPropylene. PolyPropylene was chosen for its relative inexpensiveness compared to other plastics, as well as its high impact resistance and durability. The water bottle holder on the prototype is made out of Polycarbonate and is currently manufactured by a different company. Since the initial cost to buy and begin injection molding is very high, it may be in the interest to even consider outsourcing the production of just the water bottle holder. The benefit is that the company I bought from is already working with polycarbonate and has the molds and hardware set up to create the part on a large scale.

- 1) One change in the design I would make during the manufacturing process would be to create grooves in the faces of the clamp and mount that interact with the backpack strap. These grooves would be made wide enough such that outsourced rubber inlays, thicker than the groove, could be superglued in. These rubber ridges made of silicone create a higher friction coefficient between the device and the bag strap, improving usability and stability of the product when in use. In the original design, I would have had to create a grippy surface out of the plastic itself. With these new slots, the geometry of the mounting clamps can be simplified significantly, reducing the price of the molds.

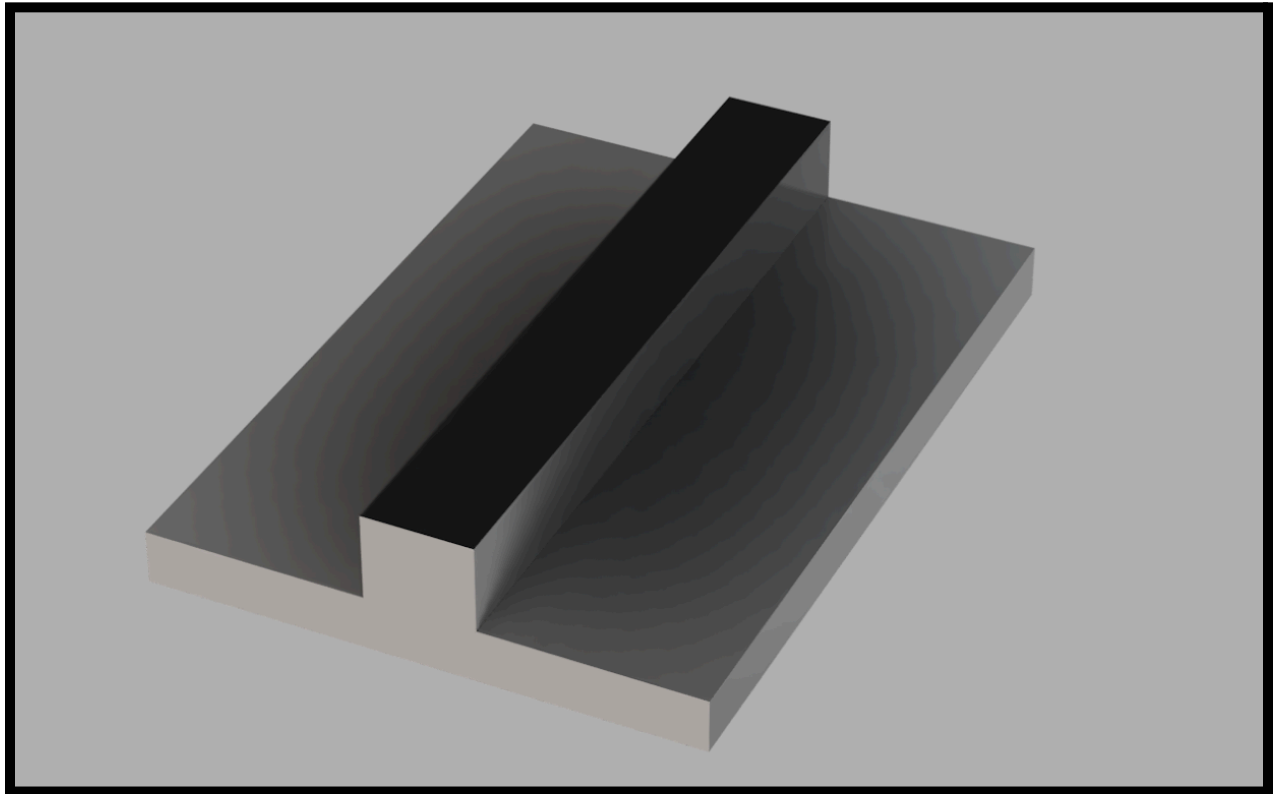


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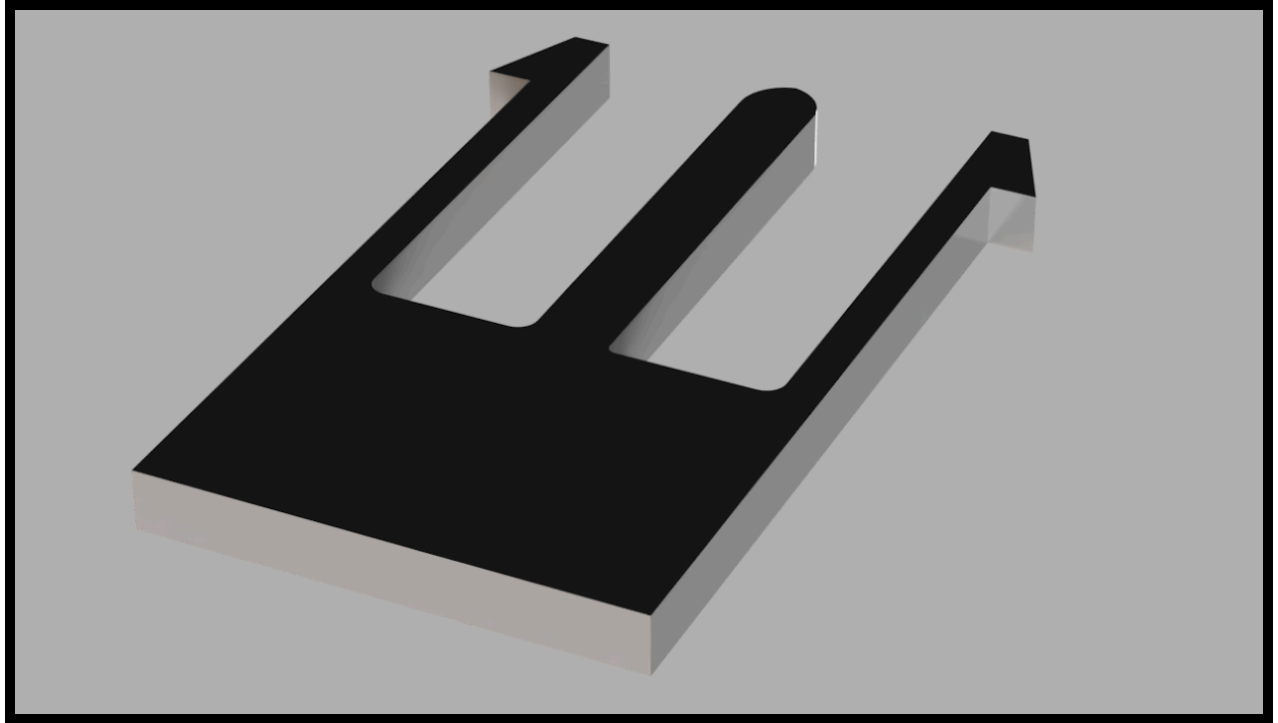
Rendered View of the rounded silicone rubber inlays

- 2) Another design change that could be made to the prototype is to convert the mounting plate and the receiver into one solid component. This results in only needing one mold for the two parts, which saves time as well as the cost of needing another mold created and fewer parts to assemble post manufacturing. To ensure injection molding goes well, I will also edit the design to make the walls of the receiver/mount uniform and thin. Rather than infilled 3d printing.
- 3) The final adjustment to the design is in the manufacturing of the clip. The clip was originally 3D printed in two separate pieces, one end being the T shape that leaves the receiver, and the other end being the buckle arms that snap the clip into place. To optimize manufacturing, I chose to again separate these two sections, choosing to laser cut the buckle arms section but keep the T body as injection molded. Benefits of this process include a less complicated mold for an otherwise complicated piece, as well as the reduced cost and increased precision of using laser cutting for the complicated 2D element (Buckle Arms).

B



T Junction Piece



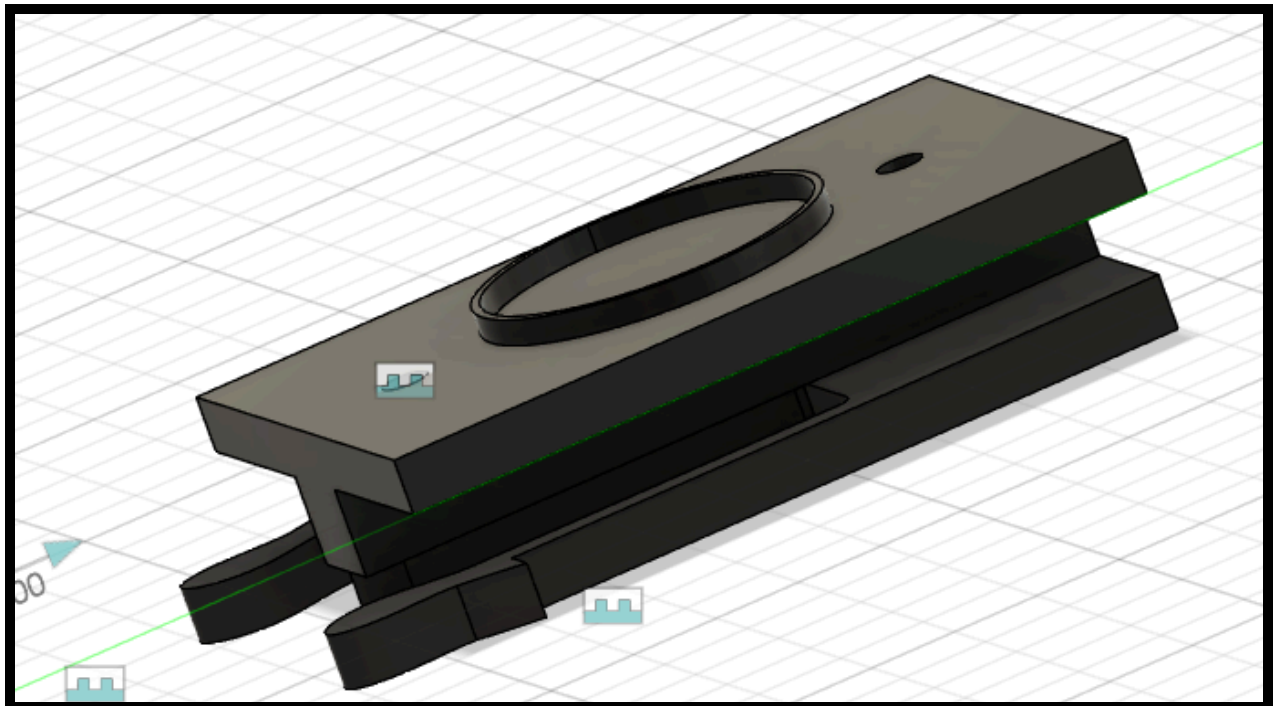
Buckle Arms Piece

Design For Assembly

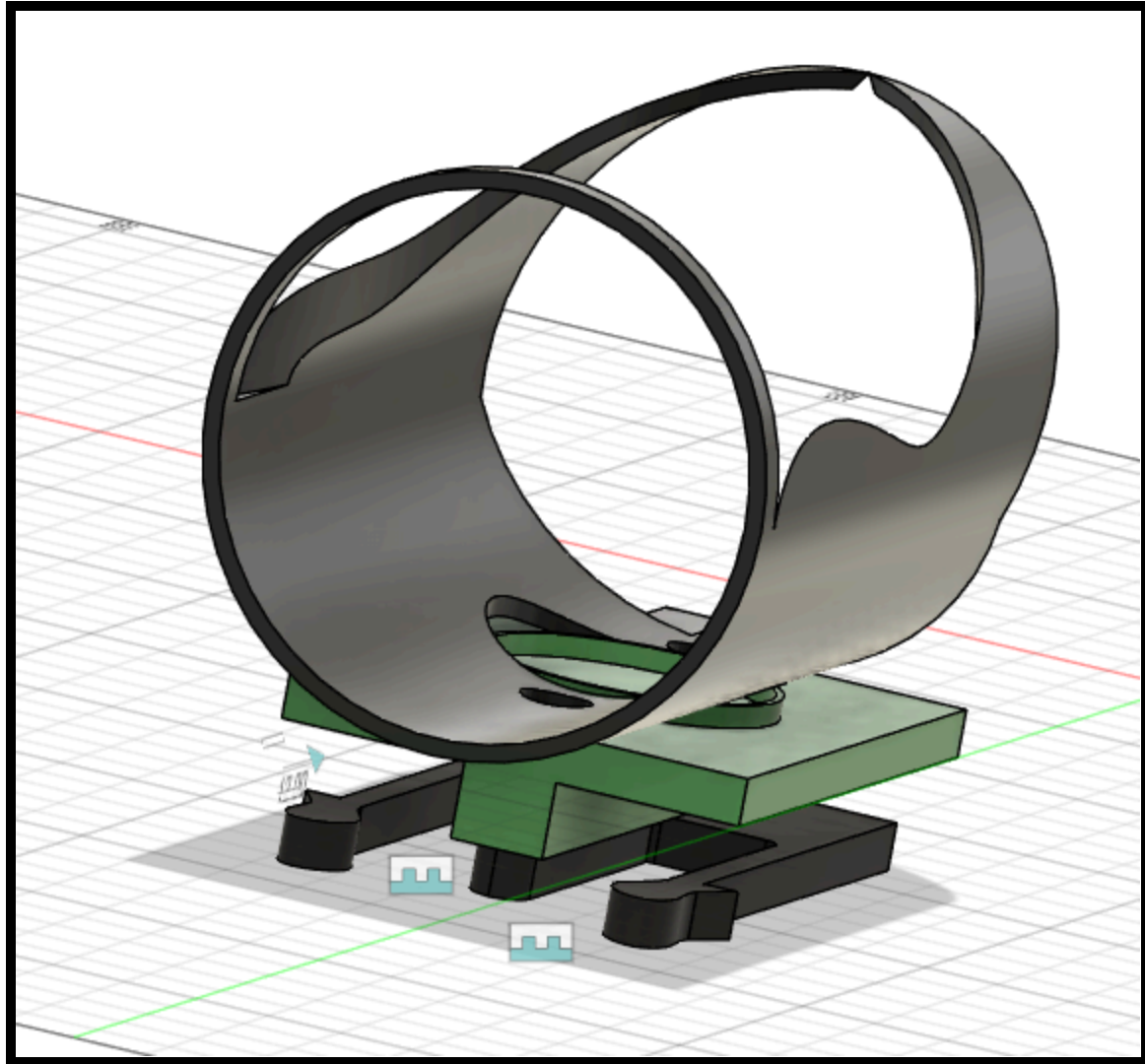
The current assembly process is rather easy. Most of the pieces are joined using super glue and the water bottle mount is connected to the buckle insert with 2 M4 screws. The clamps used for mounting are attached to the whole assembly with 2 pins.

The assembly is advantageous since the buckle insert has heat set inserts installed, so there is no need to use nuts with the screws. To attach the two together all you need is an allen key. The use of super glue also makes assembling this product a breeze since it cures so quickly. However, aligning these pieces can be tricky, making it easy to glue parts together incorrectly. Additionally the spring on the inside of the receiver is somewhat floating before being glued in place, which makes it difficult to assemble properly.

- 1) To make the assembly process easier, I would use a poka-yoke technique to align the water bottle holder with the insert. In its current orientation it is easy to install the water bottle holder upside down since there are only 2 screws, however this orientation will not work for the purposes. To fix this problem, I extruded a small ring that matches a natural gap in the water bottle holder. With this new addition, it is impossible to make the two pieces flush without having them in the right orientation, since the circular cut out is not centered on the water bottle holder.

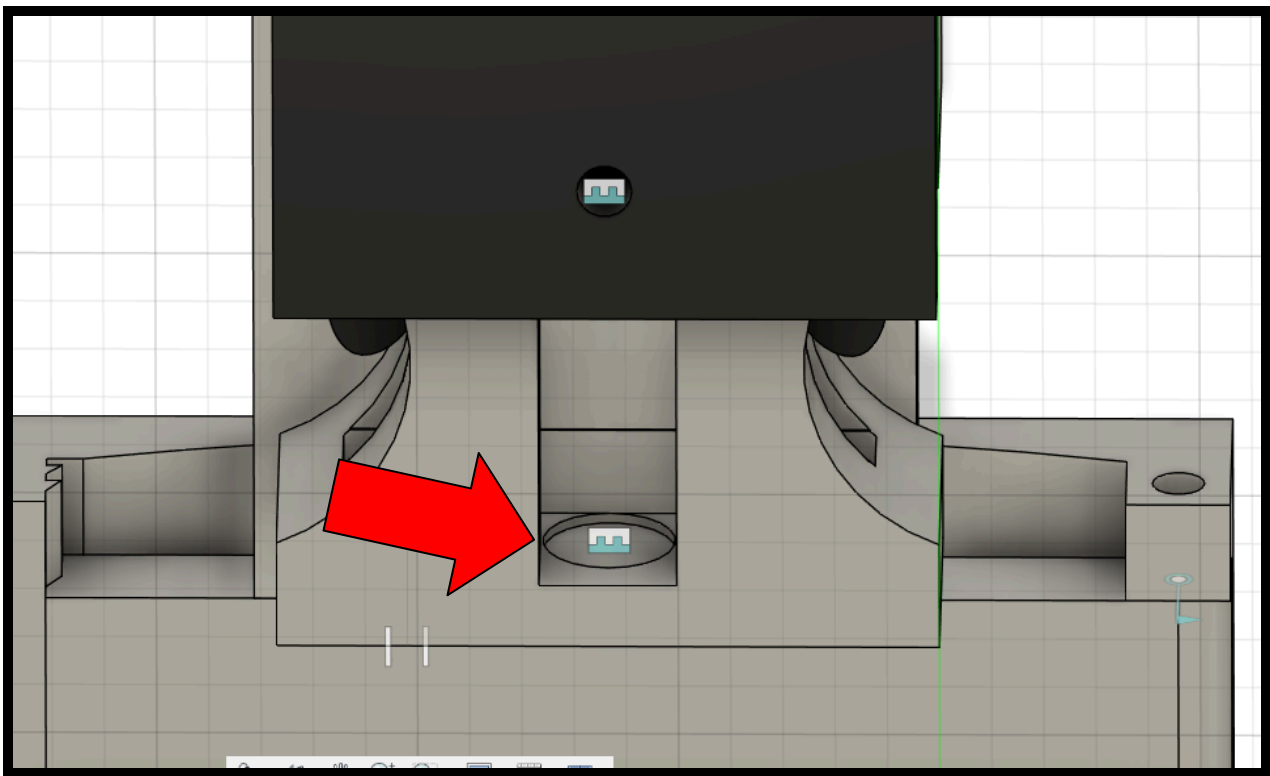


Insert piece with alignment ring

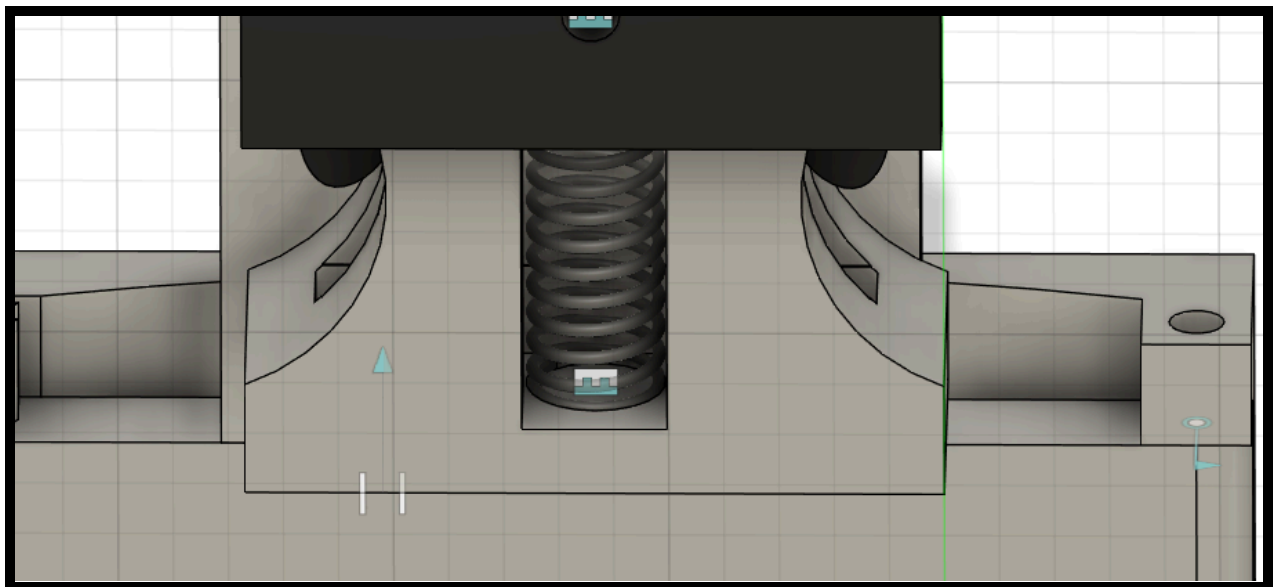


Insert piece and water bottle mount aligned

- 2) The design uses a small spring to help the water bottle pop out once the locking mechanism is released. Currently the spring is hot glued to the plastic receiver piece. To ensure that the spring is glued in the proper place, I can add a small circular extrusion to the bottom of the receiver to guide the spring into place.

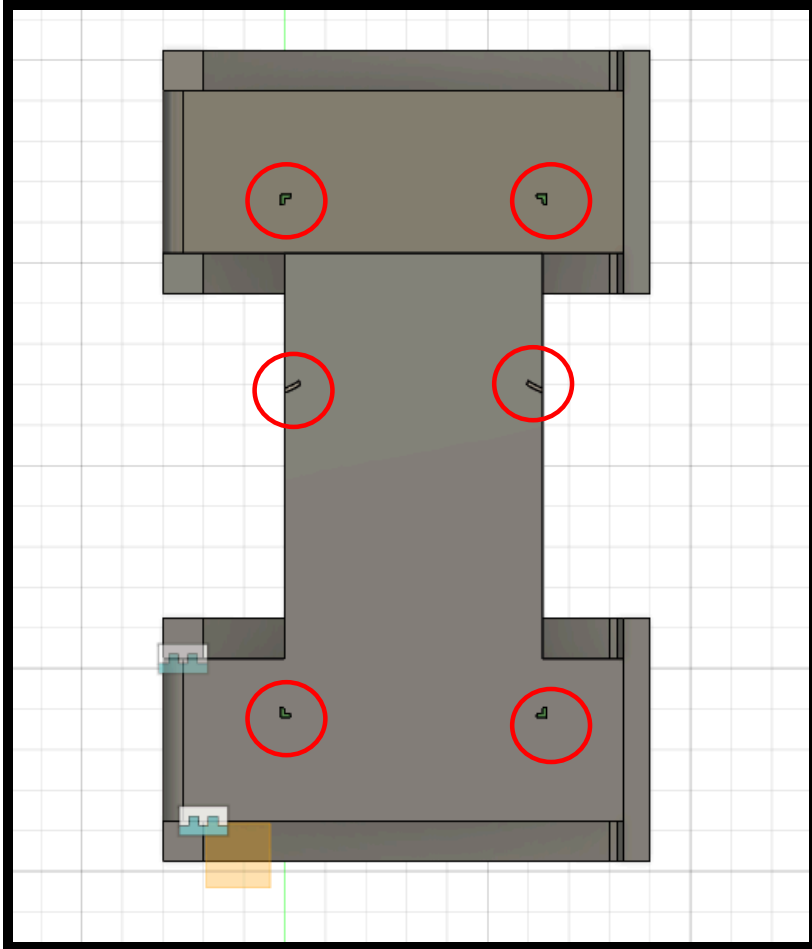


Receiver with circular spring guide

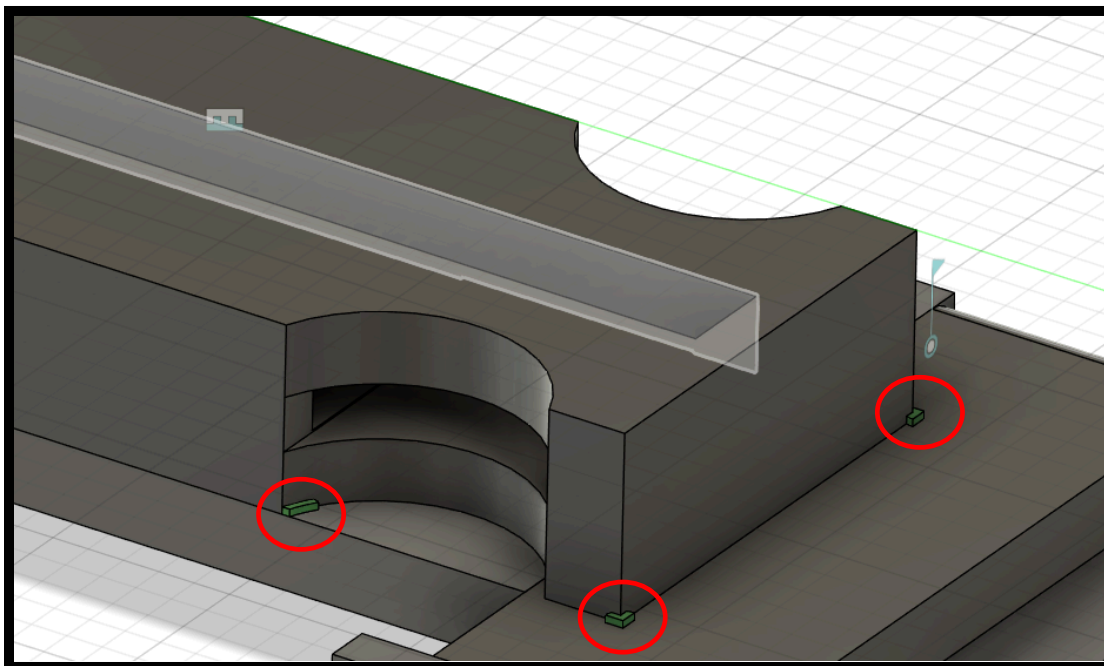


Receiver with spring installed

- 3) A similar poka-yoke technique can be used to attach the receiver to the mounting clamps. In particular, I added 6 tabs around the corners and curved parts to the mounting plate to signal exactly where the buckle receiver goes.



Mounting Plate with added tabs



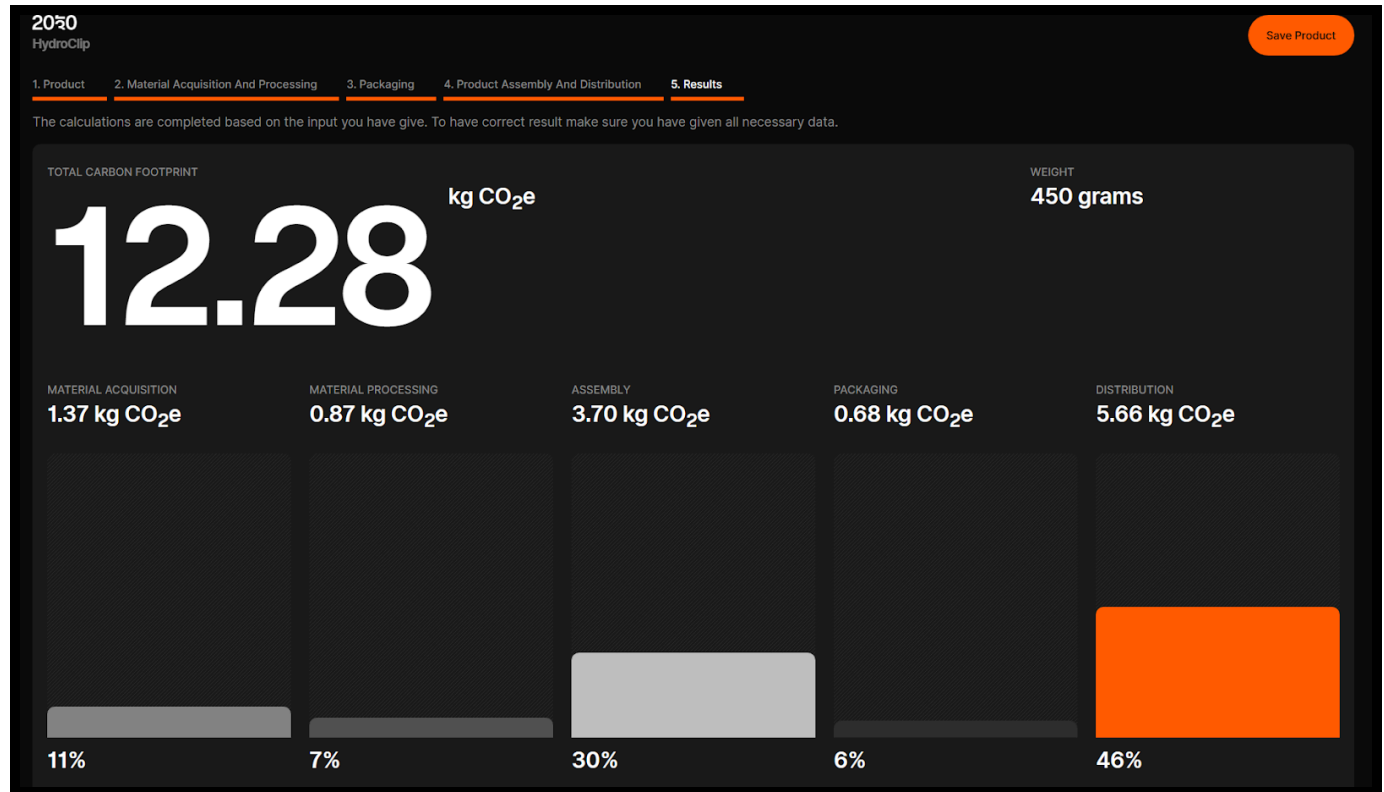
Receiver aligned with mounting plate using tabs

Design For Sustainability

The HydroClip is primarily made of (PP) Polypropylene, a common choice for many consumer products because it is lightweight, durable, and resistant to moisture. However, this material comes with significant environmental trade-offs. Polypropylene is made from petroleum, making it a non-renewable resource that has energy intensive production, contributing to a high carbon footprint. By the end of its lifecycle polypropylene can be difficult to recycle without a proper waste-stream, therefore finding its way to oceans landfills, where they can remain for decades.

The HydroClip's design consists of multiple injection-molded plastic parts that require separate molds. Although manufacturing will be energy-intensive, this approach will allow for greater precision and functionality of the product. Additionally, since the components will be permanently bonded or difficult to separate, this product will be less likely to be recycled by the end of its life. Most consumers are unlikely to spend time separating and recycling parts of the product, meaning it will most likely become mixed plastic waste.

the product is created to be durable and long-lasting, with the capability to last for several years if maintained properly. However, polypropene has the potential to degrade over time when exposed to heat, sunlight, or extreme mechanical stress which can compromise its usability.



Estimate from The 2030 Calculator

The largest contributor to the carbon footprint of the HydroClip, is primarily tied to the distribution and assembly process. Approximately half of the total footprint, 5.66 kg CO₂e (46%) is from distribution alone. This is due to the long haul-air transport required to move the product from the manufacturing facility in China to the United States for distribution which is a journey of about 7,500 miles. Air freight is also carbon intensive as it consumes fuel and emits greenhouse gases per kilogram of cargo. The second largest contributor to the carbon footprint is the assembly phase, which makes up 3.70 kg CO₂e (30%). This is due to the energy required to mold and assemble the multiple polypropylene components such as the buckle receivers, mounting plates, and clips. Each component required separate molds, significantly increasing the carbon footprint.

Potential Changes

- 1) As of now the HydroClip is made from propylene, a petroleum based plastic. Although it is durable and effective, the carbon footprint could be reduced by using a different material such as Polylactic Acid (PLA) or Recycled PET (rPET). PLA is made from renewable resources such as corn starch or sugarcane which are biodegradable meaning that they will not sit in landfills for decades. rPET

which uses recycled plastic reduces the need for raw material while still maintaining the mechanical properties necessary for HydroClip's components.

- 2) Thermoforming is a more material-efficient process that utilizes thin plastic sheets instead of thicker, solid parts used for injection molding. This reduces the raw material needed, lowering the carbon footprint of each HydroClip while also making the product lighter, which reduces emissions due to transportation. Moreover, thermoforming requires less energy, as it only requires heating a single sheet before forming it over a mold, rather than forcing molten plastic into complex shaped molds. This makes the process faster, cost-effective, and less energy intensive. Recycling thermoforming scrap is much easier due to its homogeneous plastic property.
- 3) For packaging, use lightweight, recyclable cardboard that can secure the HydroClip with minimal material. This reduces the overall weight and volume of each unit, while also cutting down on transportation emissions. Packaging can be designed with laser-cut inserts or cardboard trays that hold the product without the need for plastic inserts or excessive padding. The entire package can be processed as a single material stream which reduces the carbon footprint and simplifies recycling.

Assembly Guide



First glue the spring into the bottom of the receiver.



Then put the heat set inserts into the insert.



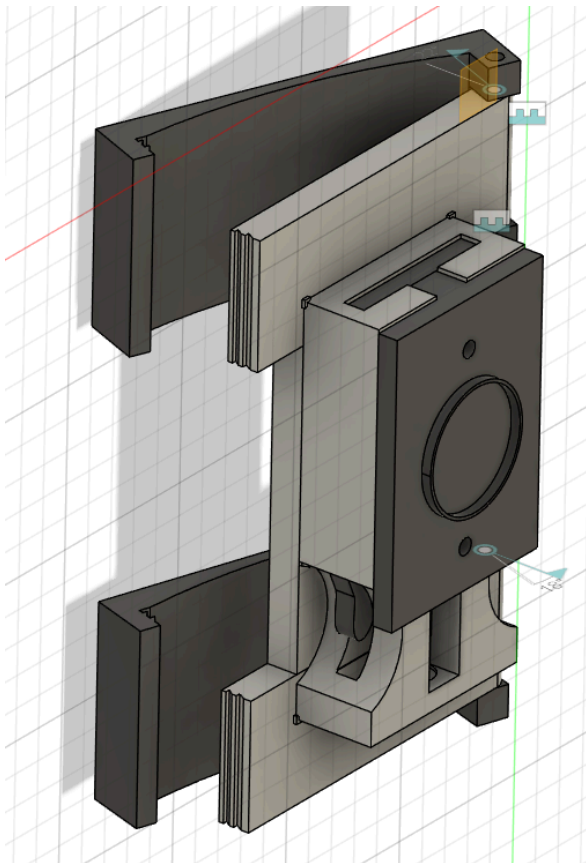
Place the water bottle holder on top of the heat set inserts.



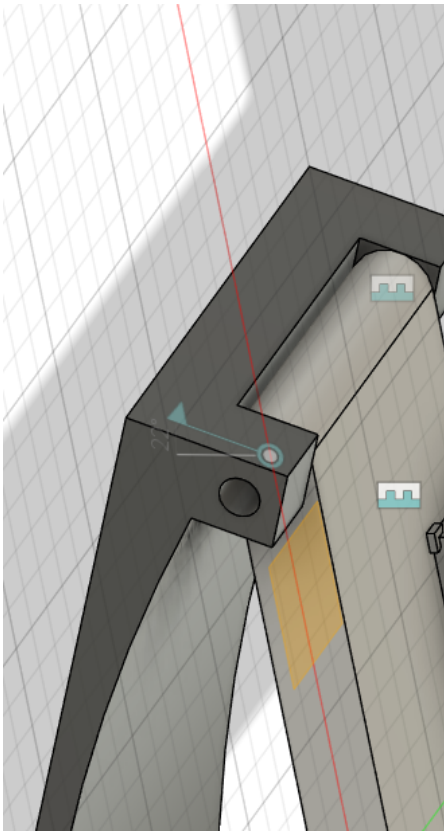
Fasten with 2 M3 screws.



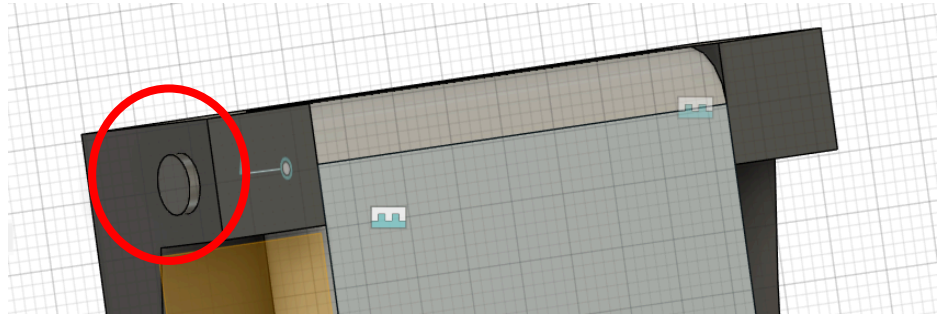
Glue hose clamps to the back and it's done!



For the improved design, the clips are custom designed, but they would be glued in the same fashion.



For this version, the assembler would need to insert pins (seen below) into each clip to facilitate the swinging motion.



Final recommendations for my product:

Based on the current design status and market potential, I would recommend extending the development cycle for the HydroClip. While the product has a solid foundation, several improvements are needed to enhance its durability, manufacturability, and overall market appeal. The extended cycle will target refining the design, optimizing manufacturing processes, and ensuring market fit through customer feedback.

In the first three weeks, I will focus on finalizing material selection and optimizing manufacturing processes. This phase is essential to choosing a more sustainable alternative to polypropylene that will meet the durability and flexibility needs of the HydroClip. I must conduct mechanical testing to confirm these materials can withstand everyday use while also minimizing environmental impact. At the same time, I will run thermoforming trials to ensure parts maintain their shape and integrity under different environmental factors.

During weeks 4-6, we will work on refining the design and prototyping. This means updating CAD models to consolidate components, like the mounting plate and receiver, into single parts to cut down on mold costs. Furthermore, poke-yoke features, such as alignment tabs, should also be integrated to reduce the possibility of assembly errors. It's also critical to gather user feedback during this phase to identify areas that need improvement before moving to full-scale production.

For weeks 7-9, the emphasis should be on user testing and market validation. During this time I will gather customer insights through interviews and testing sessions, validating HydroClip's usability, comfort, and overall value. This feedback should help dictate final design changes to improve ergonomics and usability. This is also when marketing messages should start being promoted in order to highlight HydroClip's practical advantages to build interest before the product launch.